Exc. No: 2

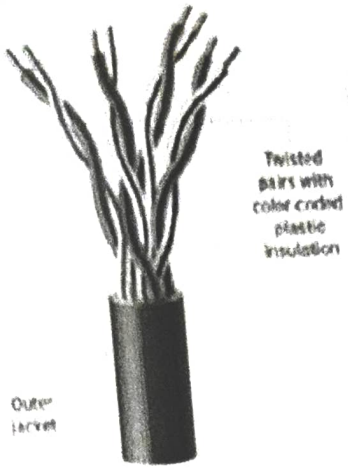
Date: 16/07/26 Study of different types of Network cables

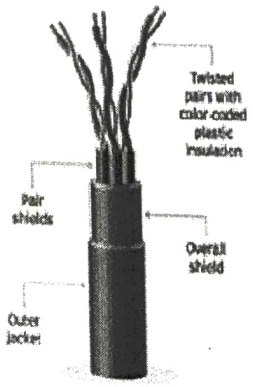
Aim:-

To study and understand different types of network cable, To make your own Ethernet cross over cable / straight cable.

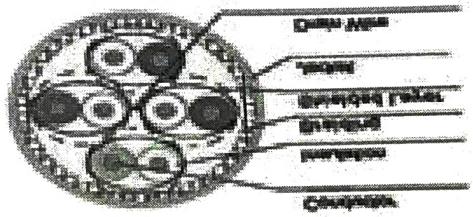
Different Types of Network cable:

1. unshielded twisted pair cable
2. shielded twisted pair cable
3. coaxial cable
4. Fibre optic cable



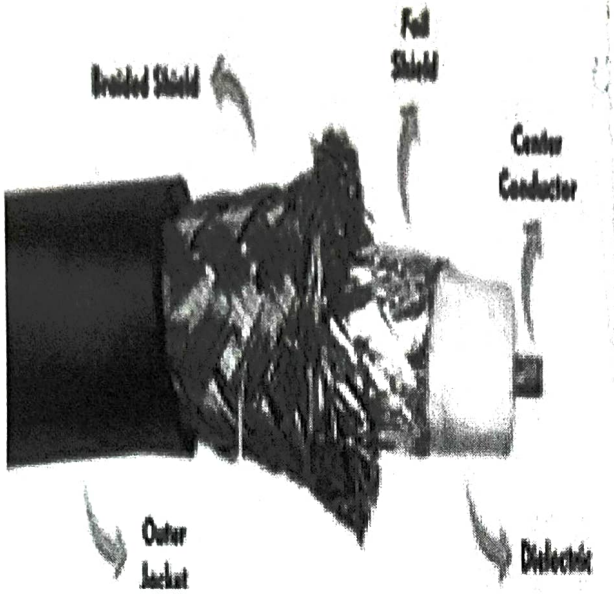


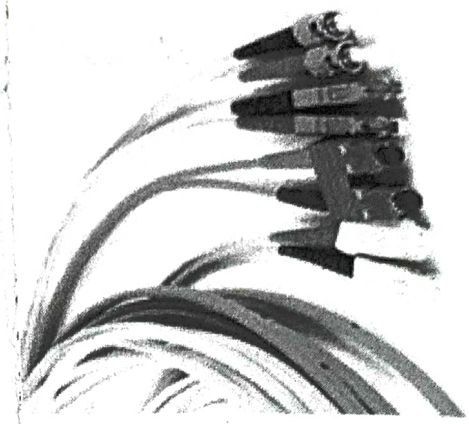
Shielded twisted pair

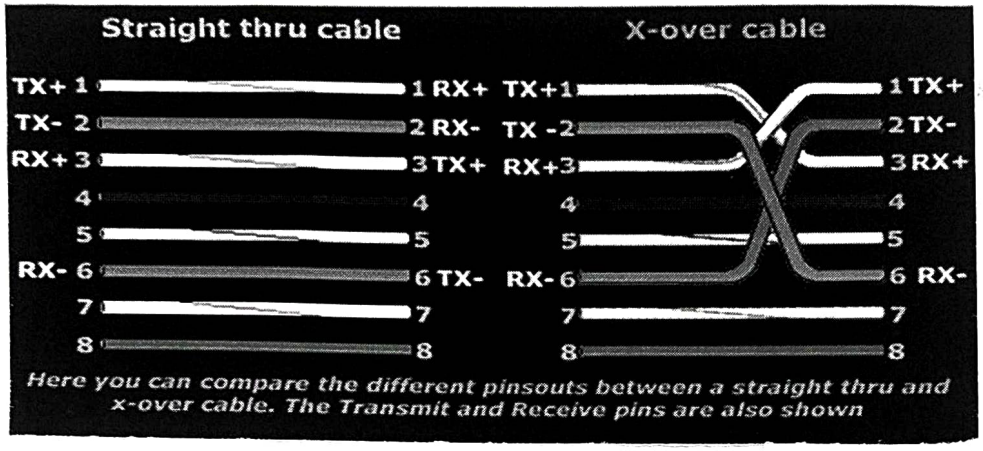




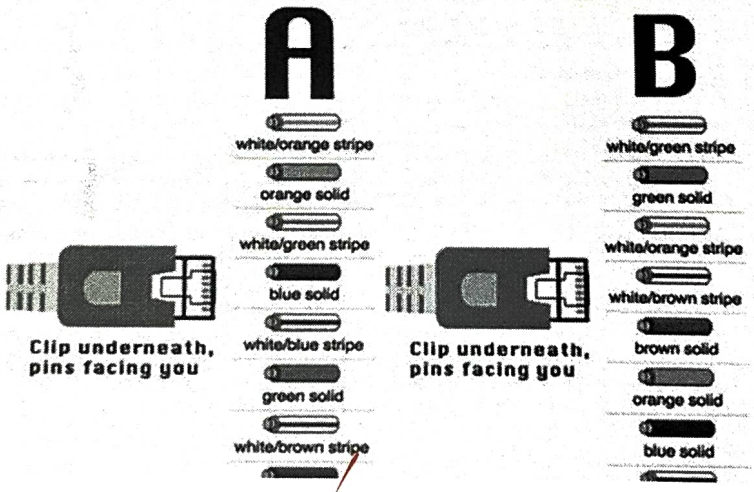
cable Type	category	maximum Data Transmission	Advantages/ Disadvantages	Applications
UTP	category 3 category 5 category 5e	10 bPS upto 100 MbPS 1 GbPS	Advantages: • cheaper in cost • Easy to install as they small overall diameter Disadvantage: • more prone to EMI	10 Base-T ethernet Fast ethernet, Gigabit ethernet
STP SSTP	category 6, 6a category 7	10 GbPS 10 GbPS	Advantages:- - shielded - Faster than UTP - less prone to noise Disadvantage:- - Expensive - Greater installion process	- Gigabit ethernet - widely used in data centres - Gigabit ethernet 10G ether net
coaxial cable	RG-6 RG-59 RG-11	10-100 MbPS	Advantages:- - High bandwith - Immune to interface - Versatile Disadvantages - cost - Size is bulky	- Speed of signal is 500m - High speed Internet connections.
fibre optics cable	singlemode multimode	100 GbPS	Advantages: - High speed - High bandwith - High Security Disadvantages: - Expensive	- maximum distance of fibre optic cable optic fast is around 100 meter







Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown



make your own Ethernet cross-over cable / straight cable :-

Tools and parts needed :-

- Ether cabling cat 5e is certified for gigabit support but cat 5. cabling works as well
- A crimping tool.
- Two RT45 plug
- optional two plug shields

step 1:

To start construction of the device, being by threading shield on to the cable.

step 2:

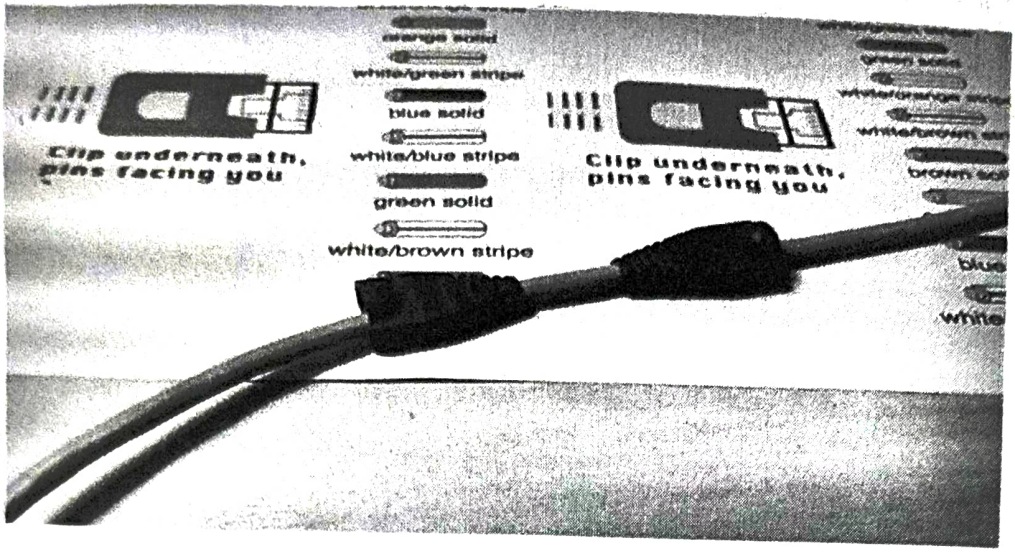
Next strip approximately 1.5 cm of the cable sheilding from both ends.

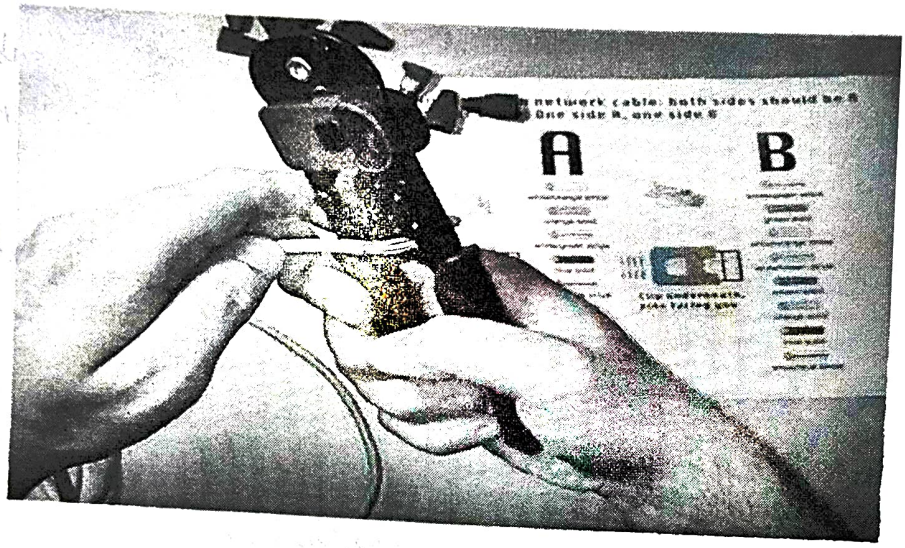
step 3:

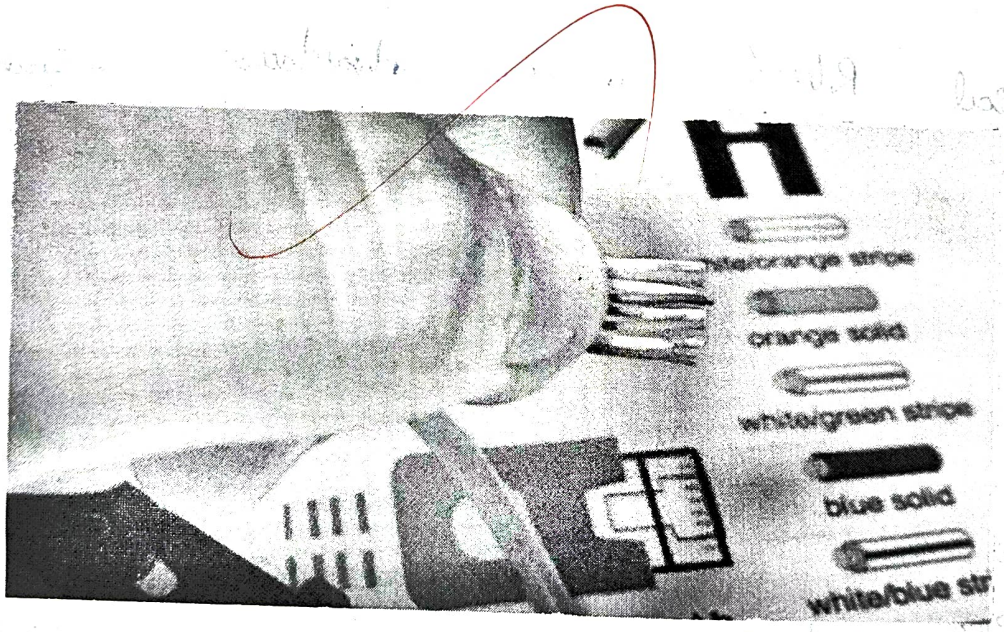
After will need to untangle the wires, there should be four twisted pairs.

step 4:

Once the order is correct, bunch them to gather in a line and if there are any that stick out farther than other Snip them back to create an even level.


white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you


A
B


B
orange solid
white/green stripe
blue solid

The different aspect is placing the RT45 plug without messing up the order.

step 5:-

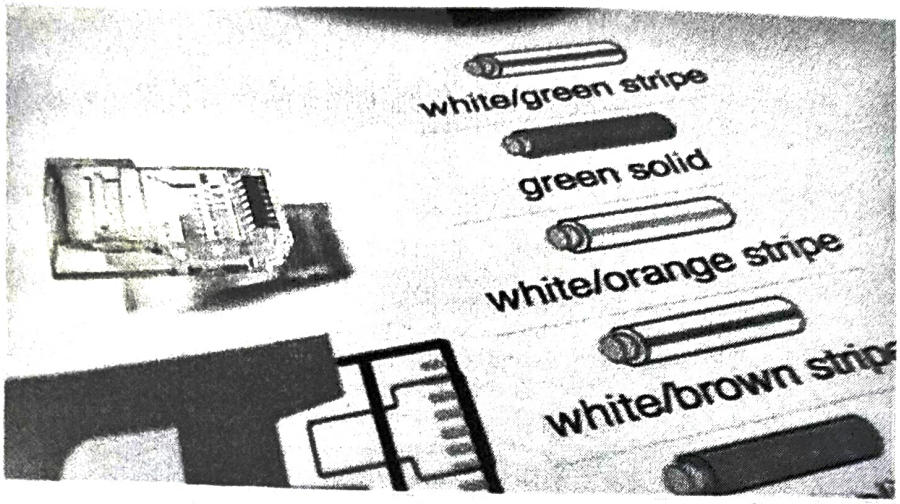
Next push the cable right in the natch at the end of the plug needs to be just over the cable shielding and if it isn't, that means that you stripped off too much shielding simply ship the cables back a little more

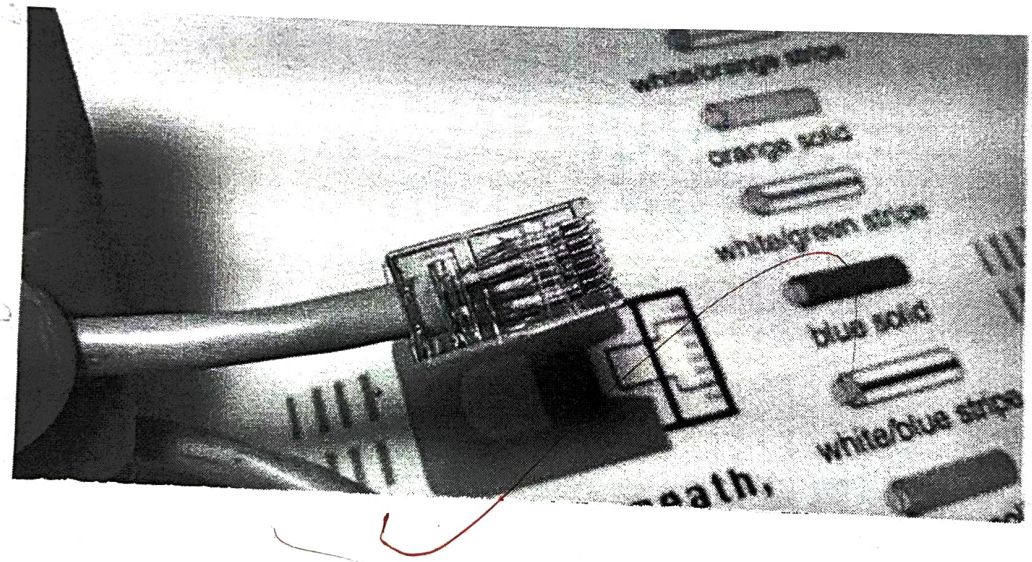
step 6:

After the wires are securely sitting inside the plug insert it into the crimping tool and push down.

step 7:-

Lastly, respect for the other and using diagram (to make a crossover cables) using diagram A (to make a straight through cable).


white/green stripe
green solid
white/orange stripe
white/brown stripe


orange solid
white/green stripe
blue solid
white/blue stripe

23/7/20

Result:-

Thus, different types of network cables were studied successfully and straight through and cross over other net were prepared and Tested successfully.